

AJINKYA GOKULE

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EDUCATION

Master of Science in Artificial Intelligence <i>Oregon State University</i> Honors Bachelor of Science in Computer Science, Minor in Economics <i>Oregon State University</i> Relevant Coursework: Data Structures and Algorithms, Deep Learning, Operating System 1 and 2, Web/Mobile/Cloud App development, Computer Security and Networking, Computer Architecture and Assembly Programming, Discrete Mathematics, Linear Algebra, Vector Calculus	GPA 4.0 <i>Expected graduation - June 2027</i>
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TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, C/C++, SQL, Bash, x86 Assembly, HTML/CSS, LaTeX AI/ML: PyTorch, Transformers, vLLM, QLoRA, GRPO, LlamaIndex, RAG, YOLO, Nelder-Mead Optimization Agents & Cloud: AWS (Bedrock, AgentCore, Lambda, S3), Strands SDK, MCP, A2A, Vercel AI SDK, OpenRouter Frameworks & Tools: React/Next.js, FastAPI, Convex, just-bash, PostgreSQL, Docker, GitHub Actions, QEMU, GDB

EXPERIENCE

Microsoft Cybersecurity Intern <i>Cloud + AI, Microsoft</i> - Architected and led a 6-engineer team building an open-source benchmark testing frontier AI coding agents against hidden prompt-injection attacks: 1,950 runs across 10 models (GPT, Claude, Gemini) in 3 safety modes - Owned end-to-end delivery of an internal AI agent that turns closed security incidents into polished, publish-ready runbooks and retrospectives enriched with Microsoft Graph context, cutting drafting time from hours to minutes - Secured three internet-exposed MCP servers that allowed unauthenticated access, owning two live incident tickets end to end, delegating fixes to engineering owners, and verifying each remediation with an auth-scanning tool - Identified ~3,000 Azure DevOps CI/CD pipelines vulnerable to supply-chain tampering via unpinned GitHub releases, engaging principal engineers to build automated CodeQL detections and prioritize fixes by blast radius	Present <i>Redmond, WA</i>
Multi-Agent Systems Developer <i>BeavsBuild at Oregon State University</i> - Lead the AWS team building a multi-agent deep research system for automated patent analysis - Integrate the emerging agent stack end-to-end: Strands SDK, AWS AgentCore, MCP tools, and A2A protocol - Drive the roadmap and cost/latency/security tradeoff decisions across the system architecture	May 2025 - Present <i>Corvallis, OR</i>
HP Software and Research Intern <i>HP Engineering Modelling and Analysis Group</i> - Led development of Physics-Informed Neural Networks to model fluid flow and heat transfer in microchips - Approximated 3D Navier-Stokes equations in semiconductor cooling, which is an unsolved physics problem - Achieved velocity profile predictions within $\pm 5\%$ margin of the analytical solution	February 2025 - September 2025 <i>Corvallis, OR</i>
AI Backend Software Engineer Intern <i>College of Business, OSU</i> - Planned and developed an interactive chatbot with a team of 6 to help 500-600 OSU business freshmen yearly - Implemented Retrieval-Augmented Generation (RAG) LLM using LlamaIndex for efficient information retrieval - Deployed a working prototype on Amazon Web Services with professor backend for uploading custom documents	April 2024 - June 2024 <i>Corvallis, OR</i>
AI Security Research Assistant <i>Secure AI Systems Lab (SAIL), OSU</i> - Researched adversarial robustness of Implicit Neural Representation and NFT-based image classifiers - Assessed robustness against PGD attacks and novel defenses on datasets like MNIST, CIFAR10, and CIFAR100	December 2023 - September 2025 <i>Corvallis, OR</i>
AI Music Research Assistant <i>SoundBendor Lab, OSU</i> - Developed unit tests to ensure consistent functionality of NoteValue class in Dr. Donnelly’s MusicAI - Tested and implemented 100 test cases for comparison, dunder, statistical and conditional operators - Presented my research in-person at the URSA Spring Symposium with a poster in the Memorial Union	November 2022 – June 2023 <i>Corvallis, OR</i>
Calculus Teaching Assistant <i>College Of Science, OSU</i> - Tutored 500+ students in a peer-to-peer environment for calculus and discrete math - Facilitated small group student learning and task accomplishment - TA’s reduced DFW (Grade D, F or withdraw) rate among students from 33% to 7%	January 2023 – April 2024 <i>Corvallis, OR</i>
Academic Learning Assistant <i>University Housing and Dining, OSU</i> - Counsel 300+ students academically through regular check-ins and resource sharing to empower them - Organize 2 events/workshops per term, fostering student development with up to 50+ attendees per event	September 2024 – Present <i>Corvallis, OR</i>
Resident Assistant <i>University Housing and Dining, OSU</i> - Supported 35 residents through regular check-ins and resource sharing to empower them - Organized 4 events per term, fostering student interaction with up to 20+ attendees per event	September 2023 – June 2024 <i>Corvallis, OR</i>

PROJECTS

(Microsoft Hackathon Winner) Trust Me Bro - AI Agent Security Benchmark [Link] - Built a prompt-injection benchmark; ran 10 frontier models through 1,950 runs across 3 safety modes - Found every model executed hidden attacks, with even the safest failing 18% of default-prompt runs - Architected a Node agent runtime around a virtual filesystem and shell, isolating runs without containers or VMs Technologies: just-bash, AI SDK, OpenRouter, Convex, Next.js, TypeScript, Tailwind CSS	June 2026
(NVIDIA Track Winner) AI Medical Bill Transparency Platform - OpenHealth [Link] - Engineered a medical bill transparency platform that parses uploaded bills and flags likely overcharges - Built a sandboxed ReAct agent with schema discovery and bounded tool-calling to prevent hallucinated prices - Developed a multi-modal vision-to-JSON pipeline using Nemotron reasoning to extract hospital bill details Technologies: AWS Lambda, Nemotron, Vercel AI SDK, TypeScript, Convex, UploadThing	May 2026
Agentic LaTeX Resume CI/CD Pipeline [Link] - Architected a pipeline where AI agents tailor resume content per job description, then build and ship via CI/CD - Encoded layout and content rules as agent-readable specs, with automated PDF verification on every release Technologies: LaTeX, GitHub Actions, latexmk, Agentic Workflows	October 2025
(NVIDIA Partnered) 3D Sound and Visual Aid - Jedi Vision [Link] - Built AI visual aid for visually impaired using computer vision and bone conduction - Deployed YOLO and depth models for real-time object detection and spatial mapping - Created 3D audio guidance via bone conduction headphones with intelligent path planning Technologies: Computer Vision, Depth Estimation, Spatial Audio	September 2025
Multi Agent Deep Research - OSU Patent Analyzer [Link] - Architected and deployed a production multi-agent deep research system automating patent prior-art analysis - Orchestrated specialized agents via A2A protocol and Strands SDK, with RAG over Bedrock Knowledge Bases - Deployed a serverless pipeline on AWS Lambda and S3, hosting agent runtimes on AgentCore Technologies: AWS AgentCore, Bedrock Knowledge Base, A2A, Strands Agent SDK, TypeScript, Lambda, S3	June 2026
(Entrepreneurial Track Winner) Student Rideshare Platform - CatchARide [Link] - Built a student rideshare platform matching drivers with spare seats to passengers on overlapping routes - Designed a custom algorithm using Route Optimization API to optimize routes between drivers and passengers - Implemented multi-authentication methods with traditional login and OAuth2.0 for secure user onboarding Technologies: Google Route Optimization API, OAuth2.0, PostgreSQL, DigitalOcean	April 2025
Full-Stack Mobile App - Campus Companion [Link] - Led an 8-person team through Agile sprints, owning task breakdown and delivery via GitHub Projects - Designed the system architecture, API interfaces, and data schemas in PlantUML before implementation - Built a realtime web-scraping microservice with BeautifulSoup backed by NoSQL caching Technologies: React Native, Expo, FastAPI, NoSQL, BeautifulSoup, PlantUML	September 2024
Secure x86 OS Implementation - JOS Operating System [Link] - Built a 32-bit x86 OS from the bootloader up: paging kernel, syscalls, preemptive scheduling, and IPC - Enforced user/kernel privilege separation and memory protection via protected-mode GDT/IDT and trap handling Technologies: C, x86 Assembly, QEMU, GDB, GNU Make, GCC, Binutils	September 2024
Lightweight RL Framework Paper - RLoRA [Link] - Co-authored and implemented a lightweight RL framework combining QLoRA and GRPO for SLMs - Stabilized training by LoftQ initialization, rank-stabilized LoRA, maintaining low KL-divergence over 500 RL steps - Reduced GPU memory usage by 26%, cut training examples by 66%, and halved compute cost while matching or surpassing SOTA with 57% pass@1 across math-reasoning benchmarks (AIME24, MATH-500, AMC23, etc) Technologies: Transformer Reinforcement Learning, QLoRA, GRPO, vLLM, Pytorch, Liteval	June 2025
(Hackathon Winner) Class Notification Extension - GetTheDamClass [Link] - A Chrome extension with 37 active users using React.js (TS) and Flask, notifying OSU students of class openings - Developed a secure backend with Docker, Flask APIs, MySQL, and Bcrypt for authentication - Reverse engineered OSU’s registration API and optimized a cron job for real-time enrollment checks Technologies: Chrome Web API, React.js, SQL, Docker, Python, Typescript	October 2024
(Hackathon Winner) AI-Generated Retro Radio Show - Backcast [Link] - Produced a 24/7 retro radio show with AI-generated music and jockey discussing historical events of each day - Implemented digital and physical interfaces, utilizing Flask on AWS EC2 and custom-modified antique radios - Engineered a robust MP3 streaming system for diverse devices and a 3JS-rendered interactive web interface Technologies: Flask, AWS EC2, OpenAI, Google Cloud TTS, PyDub, ESP32, C++/Arduino, threeJS	May 2024
(Hackathon Winner) ML Focus Extension - Focii [Link] - Built a fullstack extension using ML word embeddings to block distracting sites and boost user focus - Achieved a remarkable 75% reduction in classification error by Nelder-Mead optimization method Technologies: Transformers, Chrome Web API, Pytorch, Javascript, HTML/CSS, Node.js, Figma	October 2023
(Hackathon Winner) AI Short Video Generator - shortsGenerator [Link] - Architected a CLI tool that automates short-form video by synthesizing content from the Reddit API via OpenAI - Streamlined content pipeline to reduce video creation time from days to seconds Technologies: OpenAI API, Reddit API, Python, Bash, FFmpeg	May 2023
Multiplayer Party Game - GuessTheBeaver - Created a fullstack application inspired by the party game Spyfall for Beaver Hacks Winter 2023 - Hosted a server using Flask to enable realtime multiplayer functionality supporting upto 20 players - Designed and deployed an interactive frontend using Vanilla Javascript on Repl Technologies: Jinja, Flask, SQL, Repl, Javascript, HTML/CSS, Node.js, Figma, Git	January 2023
Macroeconomic Interactive Graph - Built an interactive Desmos tool to visualize effects of changing money supply - Connected money supply and GDP using linear and compositional functions - Used in OSU Macroeconomics to engage 200+ students per term in live demos Technologies: Desmos, Demoscript	November 2022

AWARDS AND EXTRA CURRICULARS

- 6x Hackathon Winner**, System Design Lead at OSU Product Management Club
- Musician and DJ - can play 7 different instruments, Dancer - Kinetic Kpop
- Best Teaching Assistant 2023 Award, 2x Undergraduate Research Award (URSA) Engage Recipient
- Dean’s List 7x, Drucilla Shepard Smith Award for 4.0, Honors College Differential Scholarship Recipient